Trusting the Sky: Galileo OSNMA and Cryptographic Authentication for Civil Aircraft Navigation

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Abstract:

This research examines the potential of Galileo Open Service Navigation Message Authentication (OSNMA), a cryptographic tool aimed at increasing the security of satellite navigation employed for civil aviation purposes. The relevance of this issue is caused by a steadily growing use of data provided by Global Navigation Satellite Systems (GNSSs) for determining positions, times, routes, and other aspects necessary for aircraft flights. The report concentrates on examining the specifics of authentication of Galileo navigation messages and the opportunities that OSNMA provides for enhancing anti-spoofing capabilities of civil aviation GNSS receivers. It starts by describing the problem of using manipulated GNSS data in aviation. Then, it reviews the basics of the OSNMA technique and discusses its advantages in terms of protecting receivers against spoofed information. In addition, it evaluates the drawbacks of OSNMA from the perspective of civil aviation needs. In general, OSNMA can be considered an important step towards improved security in satellite navigation; however, it has to be implemented within a comprehensive aviation safety system.

Introduction:

Global Navigation Satellite Systems have come to be critical for the safety and efficiency of civil aviation. Aircraft use data from satellite-based sources to operate flight management systems, develop flight plans, conduct surveillance and approaches, understand their position relative to terrain, and time communications. In optimal conditions, GNSS enables accurate flights, proper spacing between airplanes, economy of fuel usage, and performance-based navigation. This is why GNSS has become a silent backbone of modern aviation. Passengers cannot see its influence but it continually helps aircraft determine their current location and plan further actions.

The problem is that this dependence also makes GNSS vulnerable to external threats. The signals transmitted by satellites, which orbit the Earth thousands of kilometres away, reach ground receivers in extremely weak condition. That is why they are susceptible to radio frequency interference. The jamming process entails trying to disrupt or overwhelm the normal signal from satellites so that the receiver will not be able to receive any data from GNSS satellites. However, spoofing is more deceitful since it entails the transmission of false signals that resemble those from the satellite and which may confuse the receiver as to its location or time or position.

For aircraft, the spoofing attack might be more critical since pilots may not immediately realize the danger caused by such an attack. While jamming results in visible errors and possible loss of GNSS signal, spoofing may look less noticeable since the system still provides the wrong position. In fact, spoofing may disrupt not only the GNSS receiver itself but also related aircraft systems, such as FMS, Terrain Awareness and Warning System, ADS-B reports, navigation displays, and other avionics depending on precise positioning or timing. Hence, when the spoofing attack affects several systems simultaneously, the threat becomes broader than just providing one inaccurate information.

Modern safety guidelines demonstrate how relevant GNSS interference threats are for aviation safety. The aviation industry and regulating authorities report about the rise of jamming and spoofing in particular geographical locations, especially around the conflict regions. GNSS interference incidents can cause extra pilot workloads, misunderstandings, false warnings, navigation errors, and sometimes require route changes. All these effects make GNSS interference not just a cryptography challenge but a practical aviation safety concern as well.

However, this paper will analyze one particular solution to this problem – Galileo OSNMA (Open Service Navigation Message Authentication). OSNMA represents a way to enable message authentication of Galileo navigation messages to prove their validity and ensure they were not modified. The primary purpose of OSNMA, from this perspective, lies in enhancing spoofing detection. Thus, the central question discussed in this paper will be to what extent can OSNMA be considered useful in improving civil aircraft navigation security.

This paper will attempt to demonstrate that OSNMA is certainly a valuable innovation. At the same time, however, OSNMA cannot be used alone without other measures in place. Although OSNMA can ensure higher confidence in the authenticity of Galileo navigation messages, aviation security requires more, including proper receiver design, message authentication, multi-sensor validation, certification, training of pilots, as well as alternative means of navigation.

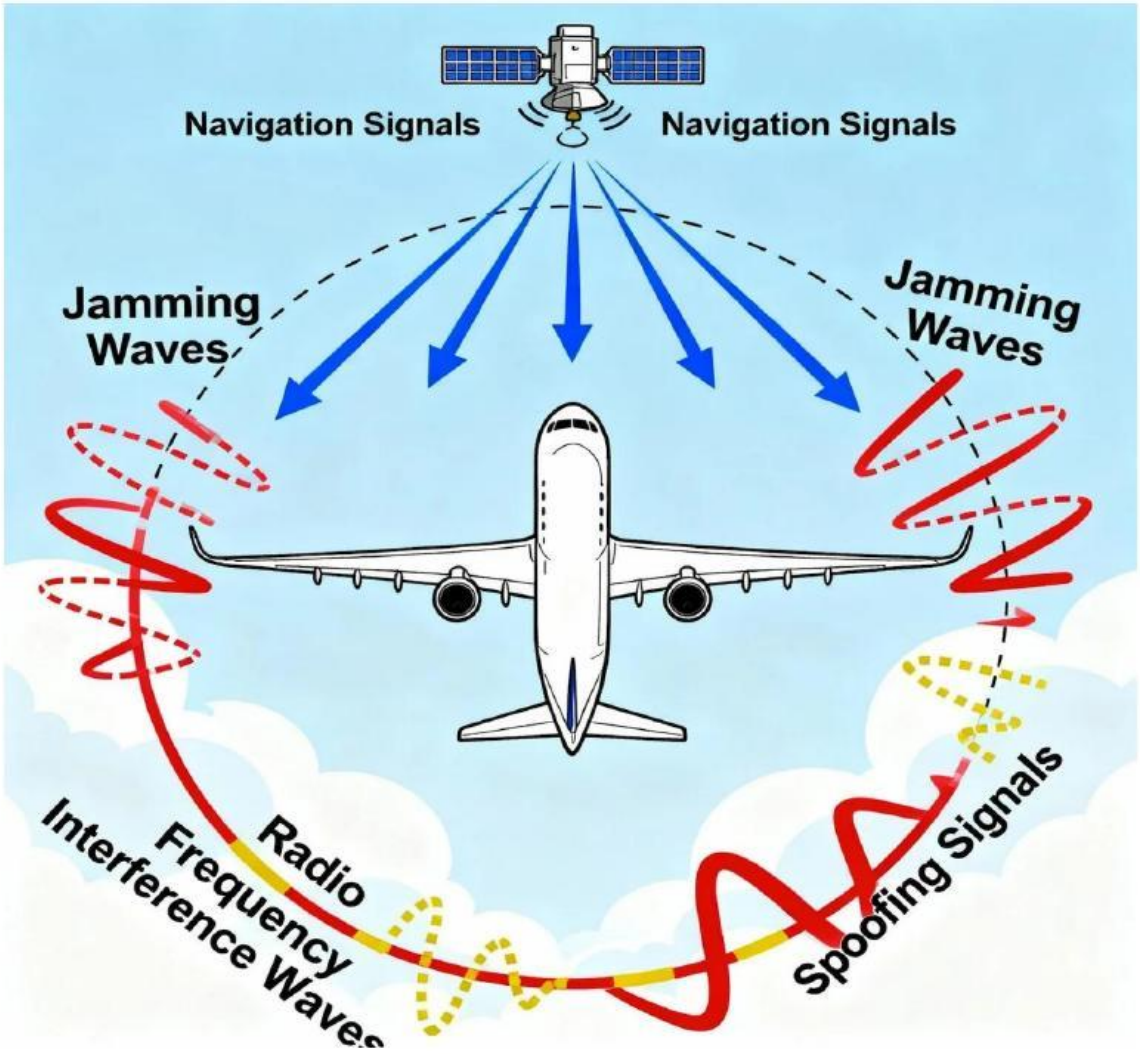


Fig. 1 GNSS Threats in Civil Aviation

The figure below shows how civil aircrafts can be threatened by various types of GNSS interference. Although GNSS signals sent to the receiver are legitimate signals sent by satellites, any interference such as GNSS jamming, spoofing, or RF interference will cause disruption or deception of the receiver. The following diagram makes us realize why GNSS security is critical to civil aviation, especially in terms of safe navigation and time calculation.



Fig. 2 GNSS Flowchart on Threats

Figure 2. Types of GNSS Threats in Civil Aviation

Figure 2 illustrates the primary types of GNSS threats in civil aviation. Jamming constitutes a form of signal denial, where interference occurs due to blockage or jamming of satellite signals, whereas spoofing is considered a signal deception threat using false coordinates, replay attacks, and clock attacks. Other environmental factors like solar flare activity, ionospheric interference, and multipath threats may affect the integrity of GNSS signals.

What Galileo OSNMA Protects (Capabilities)	What It Does Not Fully Protect (Limitations)
Navigation Data Authenticity Verifies that the navigation message (orbits, clocks, flags) originates directly from the Galileo system and has not been modified by a third party.	**RF Jamming (Denial of Service)** Cannot prevent radio frequency interference that drowns out or blocks the signal entirely, resulting in a loss of service.
Data Integrity Ensures that the content of the broadcast (such as satellite position and time parameters) remains intact and unaltered during transmission.	**Meaconing (Real-Time Replay)** Vulnerable to attacks where a valid, authentic signal is recorded and immediately rebroadcast at a different location or with very low latency.
Replay Attacks (Stored/Old Data) Protects against the rebroadcast of *old*, previously valid signals by using timestamps and cryptographic keys that expire, making the signal unpredictable over time.	**Signal Ranging/Timing Manipulation** While it secures the *data content*, it does not strictly prevent the manipulation of the signal's arrival time (ranging) if the attacker does not alter the message itself (though it makes this significantly harder to execute undetected).

Spoofing Detection Provides a "digital seal" that allows the receiver to detect and reject counterfeit or forged navigation messages, alerting the user to a potential attack.	**Service Availability in Obstructed Areas** Authentication requires successful reception of data packets; performance and availability may degrade in "urban canyons" or areas with poor satellite visibility.

Table. 1 OSNMA on what it can protect and what it cannot

Table 1. Galileo OSNMA: Strengths and Weaknesses

In Table 1 below, it is possible to see which threats OSNMA can resist, and which are out of its scope. The main advantage of OSNMA is that it improves data authentication, data integrity, replay resistance of previously captured data, and spoofing detection through the ability of users to authenticate the messages that are used for navigation. Still, OSNMA fails to offer full protection from RF jamming, replay attacks carried out in real-time, spoofing in terms of message distance, and access problems in obstructed locations.

Example of aviation application:

Imagine an aircraft using navigation services based on GNSS for landing or route navigation. The aircraft will be exposed to risks when the attacker broadcasts spoofed Galileo-based signals that confuse the receiver into making wrong estimations regarding position or time. The OSNMA technology allows the receiver to verify that the Galileo navigation message was sent from the satellite and is authenticated or that it is unauthenticated and possibly tampered with. This alone does not provide assurance that the final position estimate provided by the receiver is safe. The attack might impact signal time, pseudo range measurements, or even the receiver behaviour itself.

Description of Galileo OSNMA:

The Galileo Global Navigation Satellite System (GNSS) provides positioning, navigation, and timing services in a similar way to GPS. In this report, the focus is on Galileo Open Service, which is available for civil users and supports everyday navigation and timing functions. However, because open GNSS signals are publicly accessible, receivers cannot automatically prove that a received navigation message genuinely came from Galileo. This creates a security concern, especially for aviation, where incorrect position or timing information can affect safety-critical systems.

Galileo Open Service Navigation Message Authentication (OSNMA) addresses this issue by allowing compatible receivers to authenticate navigation data. OSNMA does not encrypt Galileo's open service signal or hide information from users. Instead, it adds cryptographic authentication material so that receivers can verify whether the navigation message is genuine and unchanged. This is particularly important for detecting spoofing attempts, where an attacker may transmit fake or modified satellite-like signals to mislead the receiver.

OSNMA is transmitted through the Galileo E1-B signal component using the I/NAV navigation message. Since GNSS bandwidth is limited, OSNMA is designed to use previously allocated but unused fields within the message structure. This allows authentication data to be added

without disrupting normal navigation services or reducing compatibility with existing Galileo Open Service operations.

The main cryptographic protocol used in OSNMA is TESLA, which stands for Timed Efficient Stream Loss-tolerant Authentication. TESLA is suitable for GNSS because satellite navigation is a one-way broadcast system: satellites transmit messages to many receivers, but receivers do not send responses back. TESLA works through delayed key disclosure. First, the satellite sends authentication tags linked to the navigation message. Later, it discloses the key needed to verify those tags. If the receiver has proper time synchronisation, it can confirm that the key was not available to attackers when the original tag was transmitted.

Message Authentication Codes (MACs) are used to protect the navigation data. A MAC is generated from the navigation message and a secret key. If either the message or the key changes, the resulting MAC will also change. After receiving the disclosed TESLA key, the receiver calculates its own MAC and compares it with the MAC received through the Galileo signal. If they match, the navigation message can be treated as authenticated. If they do not match, the message should be considered suspicious or unauthorised.

OSNMA also relies on public key mechanisms and Merkle tree validation to establish trust in the TESLA key material. Since TESLA uses symmetric keys for efficient authentication, the receiver first needs a trusted way to verify the root key and related parameters. A Merkle tree allows the receiver to validate public key information using a trusted root value and hash-based verification. If the calculated root does not match the trusted root stored by the receiver, the public key must be rejected.

The receiver verification process can be understood in two main phases: initialisation and authentication. During initialisation, the receiver obtains and verifies the necessary cryptographic material, such as public key information and the TESLA root key. This may happen as a cold, warm, or hot start depending on what material the receiver already has stored. During authentication, the receiver processes the navigation message, authentication tag, and later-disclosed TESLA key. It then validates the key chain, recomputes the MAC, and compares it with the received MAC to decide whether the message is authentic.

This design makes OSNMA well suited for GNSS broadcasting because it avoids two-way communication, fits within limited navigation message space, and allows many receivers to verify the same broadcast signal. However, its effectiveness depends heavily on correct receiver implementation. Receivers must securely store trusted material, maintain accurate time synchronisation, handle delayed key disclosure properly, and respond safely when authentication fails. In aviation, OSNMA is therefore valuable, but it must operate within a certified and reliable navigation system.

In summary, Galileo OSNMA is an authentication framework for open satellite navigation data. It uses public key validation, Merkle trees, TESLA delayed key disclosure, and MAC verification to help receivers determine whether Galileo navigation messages are genuine and unchanged. Its purpose is not to encrypt navigation data, but to improve trust in the authenticity of the received message. For civil aviation, OSNMA is an important security improvement, but it should be treated as one layer within a wider aviation safety and resilience framework.

Discussion:

Galileo OSNMA provides cryptographic authentication for navigation messages. Its main role is to help receivers determine whether a Galileo message is genuine and unchanged. This is important because traditional open GNSS signals allow receivers to calculate position and time, but they do not automatically prove that the message came from the legitimate satellite system.

OSNMA protects navigation messages through public key verification, TESLA delayed key disclosure, and Message Authentication Codes (MACs). If the verification succeeds, the receiver can have greater confidence that the message originated from Galileo and was not created or modified by an attacker. If the message is altered, the authentication tag will not match the MAC calculated by the receiver, allowing the data to be treated as suspicious.

OSNMA also improves spoofing detection at the message level. A spoofer may attempt to broadcast fake Galileo-like signals, but without valid OSNMA authentication data, a properly implemented receiver can reject or flag the message. This does not make spoofing impossible, but it makes basic attacks harder because the attacker must defeat the cryptographic authentication process as well as imitate the signal format.

However, OSNMA is not a complete solution for GNSS security. It authenticates the navigation message, but aircraft navigation also depends on signal measurements, timing, receiver algorithms, antenna behaviour, and other avionics systems. Therefore, an authenticated message does not automatically guarantee that the final aircraft position solution is safe.

One major limitation is jamming. OSNMA cannot prevent radio frequency interference because jamming blocks or overwhelms the genuine satellite signal rather than modifying the navigation message. If the receiver cannot obtain a usable signal, cryptographic authentication cannot solve the problem.

Receiver implementation is another key concern. OSNMA is only effective if receivers correctly store key material, validate public and TESLA keys, maintain time synchronisation, verify authentication tags, and respond safely when authentication fails. Poor implementation could cause receivers to misinterpret authentication status or continue using doubtful data.

Replay and relay attacks also remain possible. An attacker may rebroadcast genuine authenticated signals while manipulating timing or pseudorange measurements. This shows that OSNMA authenticates the data content, but not the full signal path or the aircraft's final position calculation.

For civil aviation, certification is also essential. Regulators and manufacturers must define how OSNMA status is displayed, how failures are handled, and how authentication interacts with systems such as SBAS, RAIM, Flight Management Systems, ADS-B, and Terrain Warning Systems.

Overall, OSNMA is a valuable security improvement because it strengthens trust in Galileo navigation messages and makes message-level spoofing harder. However, it does not stop jamming, fully prevent replay or relay attacks, fix weak receiver designs, or guarantee safe

navigation by itself. It should therefore be used as part of a wider defence-in-depth approach with receiver-level spoofing detection, sensor validation, backup navigation systems, pilot training, and aviation certification.

Recommendations:

1. Use OSNMA as a Defence-in-Depth Layer

The primary recommendation made in this report is that Galileo OSNMA should be integrated as a defensive strategy for civil aviation navigation systems, instead of viewing it as a safety system on its own. This is because OSNMA is critical in enhancing cryptography since it ensures that compatible devices can verify the authenticity of the Galileo navigation messages. Nevertheless, the safety of the aircraft navigation process cannot solely be based on the authenticity of a particular navigation message but must also factor in signal performance, receiver performance, time synchronisation, sensor validation, certification, and decision-making processes.

2. Integrate OSNMA with Aviation Integrity Monitoring

OSNMA must be integrated into the current aviation integrity solutions like SBAS, RAIM, and future ARAIM. It is crucial due to the importance of their role in the integrity monitoring of navigational information. OSNMA will allow determining the authenticity of the message received from Galileo system; however, integrity monitoring will enable evaluating the integrity and safety of navigation solution based on this navigational information. In this way, the combination of both techniques will create more robust protection than any single one of them.

3. Maintain Independent Navigation Backup Systems

The use of independent cross-checks by aircraft needs to continue. An aircraft's inertial navigation system can give a sense of position continuity in the event of problems with GPS navigation. Barometric altitude can be checked against GPS altitude for any discrepancies. Conventional navigation systems like DME, VOR, and ILS need to stay around where feasible because they will serve as backup when GNSS fails. They might seem dated, but in aviation, redundancy is not outdated; it is merely the essence of survival.

4. Strengthen Receiver-Level Spoofing Detection

The receiver should implement spoofing detection at the receiver level in addition to message authentication. This might involve testing the strength of the signal, its angle of arrival, the consistency of the pseudo range measurements, clock behaviour, multi-constellation consistency, and rapid changes in location and time information. The receiver should make distinctions between navigation messages that are authenticated, not authenticated, dubious, or unavailable. The navigation system should not continue processing doubtful data in silence.

5. Improve Receiver Protection and Key Security

Fourth, receiver protection and key security must be considered as an important factor. OSNMA requires trusted public keys, TESLA root keys, delayed key release, and MAC

authentication. Poor key management and handling by aviation receivers can make the use of OSNMA ineffective for aviation. Therefore, proper storage, protected firmware, proper update processes, and valid cryptographic material must be ensured.

6. Establish Aviation Certification Guidelines

OSNMA should be implemented via aviation certification guidelines and not merely via receiver implementation. Civil aircraft systems require reliable and certified behaviour. The aviation regulators, manufacturers, and airlines must set the guidelines on how OSNMA status will be displayed, failure recording, use of data after authentication, and OSNMA interaction with the flight management system and ADS-B among other systems. Without aviation guidelines, two receivers may handle the same OSNMA failure in different ways, which is unacceptable in aviation.

7. Support Pilot Training and ATC Procedures

The operational processes need to work in favour of technology. For instance, pilots need to undergo training on how to detect signs of GNSS spoofing and jamming which include unexpected positional changes, conflicting navigation systems, abnormal terrain alerts, ADS-B alerts, and conflicts between GNSS and non-GNSS systems. Additionally, air traffic controllers need to have guidelines on how to assist planes that experience problems due to GNSS interference by guiding them, confirming their positions via radar, and notifying other planes within the vicinity of GNSS interference.

8. Develop Future Authenticated Navigation Systems

Finally, the aviation sector needs to see OSNMA as an enabling technology towards further improvements on authenticated navigation systems. Improvements could be in multi-system authentication capabilities, enhanced protection from replay and relay type of attacks, use of authenticated ranges, and incorporation into aircraft integrity monitoring system. The importance of OSNMA is that it provides an opportunity for open GPS system users to use cryptography to challenge the authenticity of the received navigation messages. But for civil aviation purposes, it will always be safer to advocate for layered trust: message authentication, signal validation, position verification, crew training, and back up navigation capability.

Conclusion:

Galileo OSNMA is one significant advancement towards ensuring security in civil satellite navigation because it provides cryptographic authentication in an open GNSS system. Previously, civilian GNSS receivers relied on trust rather than having the capability of verifying through cryptography if the messages transmitted from the GNSS network were authentic. The introduction of OSNMA provides the ability to have cryptographic verification of Galileo's navigational signals while at the same time detecting any form of signal manipulation. This is particularly applicable to civil aviation where spoofing can provide false position and timing information.

In any case, the principal conclusion from the above-mentioned report is that OSNMA cannot be used as a complete means of ensuring aircraft safety. OSNMA can verify the message sent by GNSS satellites, yet it does not guarantee the correctness of the solution of the aircraft navigation problem. OSNMA cannot provide protection against jamming; it will also be difficult for it to protect against replay and relay attacks. In addition, OSNMA heavily relies on the work of the receiver, accurate time synchronization, proper key management, and aviation certification.

As such, the most sensible strategy would be to implement OSNMA under the concept of defence in depth. It will have to be supported by aviation integrity monitoring, spoofing detection at the receiver level, inertial navigation, barometric verification, contingency procedures, pilot training, air traffic control operations, and certification procedures. While OSNMA provides added confidence in navigation data from Galileo, civil aviation will continue to benefit from resilience strategies. Simply put, OSNMA gives planes more confidence in the information coming from the sky, while aviation will still require resilience measures throughout the process.

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Galileo OSNMA Service Definition Document, Issue 1.0, July 2025 — this is official and defines OSNMA service performance and scope.

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